

Extract, then decide: a modular pipeline of LLMs and rules to identify seizure frequency in epilepsy clinic letters

A dissertation submitted in partial fulfilment of
The requirement for the degree of
MASTER OF SCIENCE in Artificial Intelligence
in
The Queen's University of Belfast
by
Conor Brown
4th September 2026

Declaration of Academic Integrity

I certify that the submission is my own work, all sources are correctly attributed, and the contribution of any AI technologies is fully acknowledged.

I declare that I have read both the University and the School of Electronics, Electrical Engineering and Computer Science guidelines on plagiarism – <https://www.qub.ac.uk/directorates/sgc/learning/LearningResources/Plagiarism/> – and that the attached submission is my own original work. No part of it has been submitted for any other assignment and I have acknowledged in my notes and bibliography all written and electronic sources used.

I am aware of the disciplinary consequences of failing to abide and follow the School and Queen's University Regulations on Plagiarism.

Name : CONOR BROWN

Student Number: 40466218



Student's signature:

Date of submission: 4th September 2026

Extract, then decide: a modular pipeline of LLMs and rules to identify seizure frequency in epilepsy clinic letters

Abstract—Seizure frequency patterns are recorded in clinical notes stored in electronic health records (EHRs). Manual chart review is required to extract this information for research, but this is expensive and error-prone. Rules-based systems struggle with the lexical variety of seizure frequency descriptions. Large language models (LLMs) are substantially better at seizure frequency extraction, but fine-tuning a model for this task requires substantial data and computation. This paper proposes a pipeline in which one LLM call **extracts** every candidate seizure frequency event in the letter into a structured record, each with its exact evidence span, a category and a canonical label, plus a provisional answer. A second stage then **decides** the final answer from that record by applying a fixed policy. The decision was performed in two ways on the same extraction record: by rules (**Hybrid**) and by a second LLM call (**LLM-only**). Evaluation used 1,200 synthetic epilepsy letters and was compared with the benchmark previously reported for a fine-tuned LLM on a different held-out sample of the same corpus. Both **Hybrid** (Purist F1: 0.86) and **LLM-only** (0.85) exceeded the reported benchmark (0.81) and both improved on the provisional answer (0.79). Ablations of the extraction prompt show that its examples, closed label forms and evidence obligation each contribute to that result. On this synthetic benchmark, prompted extraction followed by a decision policy matched the previously reported fine-tuned result.

Index Terms—information extraction, hybrid pipeline, large language models, seizure frequency, epilepsy

I. INTRODUCTION

Epilepsy affects c. 1% of the UK population [1] and is characterised by recurrent, unprovoked seizures [2]. Seizure frequency is a key indicator of the patient’s disease burden and is commonly used in clinical care to determine the appropriate treatment and as a primary outcome in clinical trials. Seizure frequency is routinely recorded in clinic letters, unstructured narrative reports of a patient’s recent visit to the epilepsy clinic, mixed in with descriptions of past diagnoses, current symptoms and future treatment. This information can be used to conduct retrospective studies or to identify patient cohorts for clinical trials, but extracting it into that structured form requires experts to conduct costly, error-prone manual chart review.

Natural language processing (NLP) offers an automated alternative, but clinical narrative presents distinct challenges. Traditional rule-based systems provide deterministic, auditable logic valued by clinicians, but they struggle with the lexical variety and syntactic complexity of clinical frequency descriptions, which range from explicit rates (*2 to 3 seizures a week*) to relative interval counts (*4 since her last visit*) and

qualitative impressions (*frequent bouts*) [3]. Conversely, large language models (LLMs) demonstrate remarkable semantic flexibility in zero- and few-shot clinical information extraction [11], [12]. However, unconstrained LLMs introduce substantial clinical risks: they operate as black boxes, remain prone to hallucinations or ungrounded assertions [5], and existing state-of-the-art results have relied on fine-tuning on large collections of expert-annotated letters [3].

Seizure frequency extraction is framed as a single-label extraction task in prior work [3], but it consists of multiple sub-tasks. A clinic letter can describe several seizure events of different types and on different occasions, so the system must first **extract** every candidate fact with its evidence, then **decide** which candidate, or which combination of them, best represents the patient’s current seizure frequency burden. Which to keep is a clinical policy that varies between users. Keeping the two stages apart captures more evidence and makes the final answer explainable, as the end user can see the candidates, the policy and the label it produced.

The primary objective of this paper is to evaluate whether a two-stage extraction pipeline, where the LLM first extracts every candidate and then decides which to keep, can extract seizure frequency patterns from epilepsy letters with comparable accuracy to past benchmarks. [3]. The secondary objective is to evaluate whether the **decide** stage improves on the provisional answer from the initial **extract** stage. The **decide** stage is implemented as both a second LLM call (**LLM-only**) and a rules-based system (**Hybrid**) and evaluation will determine whether a **Hybrid** system offers complementary benefits that an LLM-only system cannot. If both objectives are met, the paper contributes a new method for seizure frequency extraction.

To answer these questions, the pipeline was evaluated on 1,200 synthetic epilepsy clinic letters produced by Gan *et al.* [3], each with a single expert-annotated label for the patient’s current seizure frequency pattern. Results are compared with the benchmark Gan *et al.* report for their fine-tuned LLM. The primary evaluation metric is the F1 score. The two papers use different held-out samples from the same synthetic corpus; Section IV-A reports the uncertainty this introduces.

The rest of the paper is organised as follows. First, the literature review summarises previous methods used to extract seizure frequency patterns from epilepsy letters, compares conceptual framings on the different sub-tasks of clinical IE and establishes how this method is different. Next, the methods

section describes the dataset used for evaluation, the overall architecture of the pipeline and the specific implementation details of the LLM and rules. Next, the results section presents the results of the experiments, comparing the method against past benchmarks and examining its strengths and weaknesses. Next, the discussion section presents implications of the results along with the limitations of the work and ethical considerations. Finally, the conclusion section summarises the key findings and highlights opportunities for future research.

II. LITERATURE REVIEW

The best approach to extract seizure frequency information from epilepsy clinic letters is an active area of research, with experiments using rules-based methods [4], [6], transformers-based methods [3], [5], [7], [8] and a combination of both [9], [10] all published this decade. Rules-based methods struggled with the lexical variety of seizure frequency patterns. A key challenge with both rules-based methods and BERT-based methods was their failure to generalise to letters from a different hospital. [6], [10]. LLMs were the first method to demonstrate strong generalisation to a new dataset [3], but this was achieved by fine-tuning a model on over 1,000 synthetic letters. This paper explores whether similar performance can be achieved with a two-stage extraction pipeline.

Clinical IE has been defined as the automatic extraction and encoding of clinical information from text [13]: recognising a mention and normalising it into a standard form [14]. Modern LLMs do both in one step when asked for a normalised label such as *4 per month* from a raw statement such as *about once a week*. A further step exists in many clinical extraction pipelines but goes unmentioned: deciding between candidates. In a clinic letter with several seizure frequency statements, producing a single label for the patient’s current status requires the system both to extract the relevant statements and to decide which one best answers the question. Prior rules-based systems made this a distinct stage [15], LLM pipelines instead make the decision inside the extraction prompt. This paper explores whether separating *extract* from *decide* in a two-stage pipeline can deliver comparable performance while preserving more of the evidence for downstream use.

Academic papers tend to focus exclusively on benchmark accuracy while practitioners tend to value transparency, configurability and efficiency at the cost of some accuracy [15], [16]. Model interpretability is also a critical factor clinical settings [14]. Clinicians are trained to use deductive logic to draw conclusions and they expect systems to expose transparent, interpretable evidence that they can reliably use to make decisions that demand high accountability. Past work utilises evidence spans as a training input [3] for fine-tuning or as data exhaust as part of using a rule-based method [4]. This paper explores whether preserving a transparent evidence chain from exact text to final answer can make the output both more interpretable and more accurate.

Thank you for asking me to review your patient. I am concerned that her epilepsy has worsened in recent months, with a clear pattern of deterioration. She and her partner report increasing clusters of events correlating with the perimenstrual period, suggesting a catamenial component. In particular, she describes brief focal episodes with preserved awareness that tend to occur as she is settling to sleep, at the point of drifting off, now happening on most nights of the week. These events are accompanied by a rising epigastric sensation, facial pallor, and right-hand automatisms lasting 30–60 seconds, followed by fatigue and headache. In addition, she has had three generalised tonic-clonic seizures in the last six weeks, two of which occurred during the same perimenstrual window.

Fig. 1: An example clinic letter with two distinct seizure frequency patterns highlighted as potential candidates.

TABLE I: Dataset summary

		Development	Test
Number of letters		750	450
Seizure frequency	Frequent	51%	50%
	Infrequent	17%	18%
	Unknown	17%	17%
	Seizure free	15%	15%

III. METHODS

A. Data

The dataset is a collection of epilepsy clinic letters which record a UK patient’s consultation with an epilepsy specialist, often a neurologist or specialist nurse [3]. The NHS requires that clinic letters are sent to the patient’s GP after all outpatient visits such as these. They are often the only record of a patient’s diagnoses, treatments and outcomes in neurological disorders, and they are written in narrative form. An example clinic letter is shown in Figure 1.

Gan *et al.* [3] produced this dataset of synthetic letters modelled on real clinic letters from King’s College London hospital. The real letters were used to create 10 template letters and they were combined with 500 short descriptions of seizure frequency patterns produced by GPT-5 to create the full set of 15,099 letters. A sample of 1,200 letters of was used in this paper. Trained experts annotated each letter with a single gold-standard label representing the patient’s current seizure frequency status.

Minimal preprocessing was applied: letters were cleaned only to parse quotes, new lines, tabs, the $\leq$ sign and nulls, and were ingested in full with no tokenisation, section-splitting or truncation.

The dataset was split into a development set of 750 letters and a test set of 450 letters, stratified by their seizure frequency category (frequent, infrequent, unknown, seizure free) to balance the distribution across both splits. The exact breakdown can be seen in Table I.

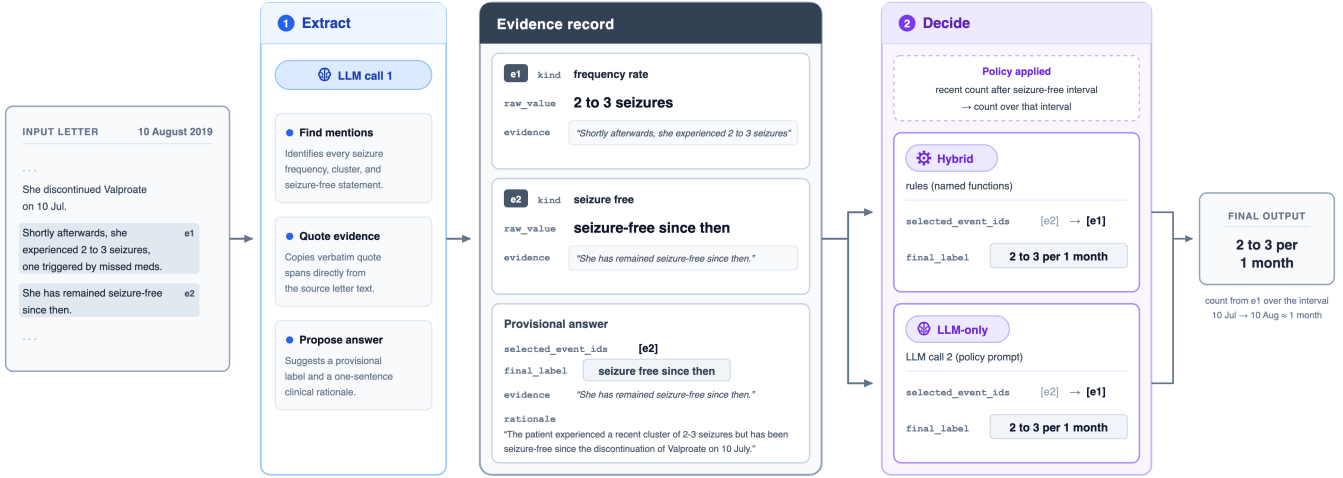


Fig. 2: The pipeline on a development-set letter. The letter narrows to a record and the record to a label: on the development set the extraction call turns a letter of about 400 words into a record of 2.2 events on average (median 2, range 1–6) holding 22 quoted words, and decide returns a label of about four words.

B. Architecture

The pipeline is composed of two stages, *extract* and *decide*. The pipeline receives an the epilepsy letter in unstructured text as input and the output is a structured evidence record in JSON. The *extract* stage builds the evidence record and the *decide* stage makes decisions based on it. Figure 2 traces the path of a single letter through the pipeline.

The evidence record contains the final answer to the question: *what is the patient’s current seizure frequency burden?* Crucially, it also contains the evidence gathered to determine that answer: multiple seizure frequency statements, exact citations from the next plus normalised labels, and the rationale for choosing that particular seizure frequency pattern.

Extract is an LLM call. It receives the full letter as input along with a detailed set of prompt instructions. The prompt instructions describe the seizure frequency task, the required output schema and examples of how to complete the task. It generates the evidence record illustrated in Table II.

Decide can be implemented as a second LLM call (LLM only or a set of rules (Hybrid)). Both pipelines share the same *extract* stage, operate on the same structured evidence, and apply the same set of procedures. Direct comparisons between the two methods measure how effectively they apply the decision policy.

C. Extract

Extract is the most consequential stage. It ingests the full letter, generates the evidence record and provides the provisional answer. The prompt has four components:

- 1) *Instructions*: find every seizure frequency statement, represent each as an event, assign it one of the six categories in Table II, and propose a provisional answer for the patient’s current seizure frequency.

TABLE II: The evidence record: output of *extract*, input to *decide*

Field	Content	Example Value
<i>One event per seizure frequency statement in the letter</i>		
event_id	Identifier used by decide	e1
kind	Frequency rate, cluster frequency, seizure free, last event only, unknown frequency, no reference	frequency rate
raw_value	The stated rate, duration or last-event statement	2 to 3 seizures
evidence	Exact quote from the letter	Shortly afterwards, she experienced 2 to 3 seizures
<i>The provisional answer, one per letter</i>		
selected_event_ids	The event or events answered from	[e2]
final_label	The label	seizure free since then
evidence	Exact quote supporting the answer	She has remained seizure-free since then
rationale	One sentence of reasoning	... has been seizure-free since the discontinuation of Valproate

- 2) *Examples* of how events should be represented.
- 3) *Allowed labels*: a closed list of the canonical label forms used by the gold standard (*N per unit*, *N per N unit*, *seizure free for N unit*, etc.). Each allowed label comes with examples and rules for writing them: digits not words, a bound such as *up to four per day* flattened to *4 per day*, etc.
- 4) An *evidence obligation*: the *evidence* fields of Table II, with the instruction that each must be an exact

TABLE III: Decision policy: when the final answer differs from the provisional answer

The record contains	Final answer
An overall count and a breakdown by seizure type*	The overall count
A seizure-free statement while other seizure-like events continue*	The continuing events
A year-to-date total and a typical rate	The typical rate
Counts for named months	Total over those months
Seizures on separate dates or months	Count over that span
A recent count after a stated seizure-free interval	Count over that interval

quote from the letter.

The model therefore does three things in one call: it finds the relevant statements and quotes them, it normalises each into an allowed value, and it chooses among the events. Each of these prompt components are removed in ablations in the results section to evaluate their impact. The full prompt, category definitions and label forms are in the supporting materials.

The type of data captured in the `extract` stage is shown in Table II. The `final_label` is the provisional answer provided by the LLM and it serves as a single prompt baseline in this paper. That single label is directly equivalent to the full output by single-stage LLM pipelines in past papers [3]. What distinguishes this pipeline from previous work is that it also records every other seizure frequency pattern in the text.

D. Decide

Decide is a smaller and narrower task. Its input is the evidence record; it does not see the full letter and cannot generate new evidence. It keeps the provisional answer unless a rule in fixed decision policy applies, in which case it either chooses a different event from the record or writes a new label from several of them. Table III lists the policy. In Figure 2 the initial extraction proposed *seizure free* based on the recent seizure free period, but the policy overrides this and prefers the cluster to 2-3 seizures just before it.

In `Hybrid`, each rule is a named function with `regex` matching and if-then logic that reads the record. A rule either selects a different already-extracted event or derives a new label from several events with date and numeric logic.

In `LLM-only`, each rule is a written instruction with one worked example: a small record, the first choice, and the expected answer. The prompt tells the model to keep the first choice unless a row applies, to write a new label only when no single event is the answer, and never to add events or quotes. It returns the selection block of Table II. The two executors differ in how the guards are expressed (code against prose), in how a pattern is recognised (a fixed expression against the model’s reading of the quotes), and in determinism: the rules always return the same answer for the same record.

E. Evaluation

F1 is the standard metric used to report results of seizure frequency extraction, but authors differ in the granularity at which a correct frequency label is defined. To allow comparison with

TABLE IV: Comparison with previously reported results

Author	Year	Method	Purist F1	Pragmatic
Holgate et al.	2024	Prompted LLM	-	0.73
Holgate et al.	2025	Fine-tuned LLM	0.68	0.81
Gan et al.	2026	Fine-tuned LLM	0.81	0.85
This paper	2026	Hybrid	0.86	0.88

Each row was evaluated on a different held-out sample; only Gan et al. and this paper share a corpus and metric definitions.

Gan *et al.* [3], this paper adopts their two evaluation categories: a fine-grained `purist` category and a coarser `pragmatic` category. If the gold label is *twice a week* and the prediction is *once a week*, that is a correct `pragmatic` answer, since both fall in the `frequent` category, but an incorrect `purist` answer. Table V lists the categories in each.

Micro-F1 is reported for both because the categories are badly imbalanced (some have over 100 letters, some fewer than 10; Table V) and micro-F1 gives each letter equal weight. Each letter has exactly one gold label and one prediction, so micro-F1 here equals the share of letters answered correctly. The more precise `purist` F1 is the primary metric.

All prompts, schema and rules were developed on the 750-letter development set, using qualitative error analysis to find failure modes and test changes. The pipeline is evaluated on the held-out test set of 450 letters. Results are compared against the evaluation set labelled Synthetic (1,166) in Gan *et al.* [3], a different held-out sample from the same synthetic corpus, distinct from their fine-tuning data.

The primary model is Gemini 3.7 Flash. It is the only model used for the prompt ablations and for the temperature and thinking-level ablations. Six models were evaluated as the extraction call in `Hybrid` (Section IV-D). Elsewhere, `Hybrid` results are from Gemini at temperature 0 with low thinking effort.

IV. RESULTS

A. Benchmark Performance

`Hybrid` achieves a `purist` F1 of 0.86, above the reported benchmark of 0.81 for the fine-tuned LLM [3]. `LLM-only` also exceeds it (0.85) The paper used a different sample from the same corpus so comparisons are limited, but the results indicate that the two-stage pipeline produces comparable results to a fine-tuned model. The first LLM call alone, scored on its provisional answer, reaches 0.79, close to the benchmark. Rules applied in the `decide` stage add +0.07 in `Hybrid`; the second LLM call adds +0.06 in `LLM-only`.

The pipeline is particularly effective at extracting high-frequency events. Gan *et al.* report unknown as their highest scoring category (0.89), with `daily` (0.70) and `more than weekly` (0.68) much lower. In this pipeline, `daily` (0.92) and `more than weekly` (0.91) are among the best performing categories, well above unknown (0.74). The full breakdown across `purist` and `pragmatic` categories is in Table V.

TABLE V: Hybrid: Purist and Pragmatic categories

Purist	n	Purist F1	Pragmatic F1
Daily	41	0.92	Frequent: 0.95
More than weekly, less than daily	123	0.91	
Once a week	5	0.89	
More than monthly, less than weekly	58	0.90	
Once a month	18	0.94	Infrequent: 0.82
More than every 6 months, less than monthly	55	0.81	
Once every 6 months	1	0.00	
Less than once every 6 months	6	0.60	
Unknown	76	0.74	Unknown: 0.74
Seizure free	67	0.90	Seizure free: 0.90

Rows with $n < 10$ shown in grey.

TABLE VI: Purist F1 after each stage, by method

Method	Provisional answer (extract)	Final answer (decide)
Hybrid	0.79	0.86
LLM-only	0.79	0.85

B. Provisional and Final Answers

Table VI reports `purist` F1 at two stages: the provisional answer after `extract`, and the final answer after `decide`, for each method. The provisional answer is the LLM’s selection in the `extract` prompt. The difference between the columns is the value `decide` adds by reviewing the evidence and applying the decision policy outlined in Table III.

The provisional answer from the single `extract` stage is 0.79. This is equivalent to the entire pipeline in most prior LLM studies, which ask the model to extract and decide in one prompt. Both methods then improve on that shared provisional answer at `decide`: rules add +0.07 and the second LLM call adds +0.06. This shows the `decide` stage has a substantial effect, and this task can be performed by rules and a second LLM call equally effectively.

C. Extraction Prompt Ablations

Table VII removes one ingredient of the extraction prompt at a time and evaluates the `Hybrid` pipeline at both stages, so the effect of each ingredient can be seen on the provisional answer and after rules have applied the policy. Removing the worked examples, the closed list of allowed label forms, or the evidence obligation each lowers the provisional answer by -0.02 and the final answer by -0.04. Their individual effect is roughly equal. What this shows is while each component has some effect, the prompt design is fairly robust to any one change.

The result for the evidence obligation is of particular interest. Examples and label forms are designed to help the LLM choose the right answer by describing what the right answer looks like. The evidence obligation is different; it is an additional layer of task complexity to include the exact evidence quotation along with the answer label. Including the

TABLE VII: Extraction prompt ablations (Hybrid)

Extraction prompt	Provisional answer	Final answer
Full prompt	0.79	0.86
Without examples	0.77	0.82
Without evidence obligation	0.77	0.82
Without closed allowed-label forms	0.77	0.81

Each row changes only the extraction prompt; the same rules apply the policy at `decide`. Effects are not additive.

evidence helps at the `decide` stage because the rules can use the exact quotes to determine the correct answer even if the LLM provides the wrong label. However, it also appears to help in the initial `extract` phase (+0.02) by encouraging the LLM to ground its answer in the evidence. This shows that there is no trade-off between schema complexity and accuracy; a more complex schema with more stringent requirements produced more evidence and more accurate answers.

D. Model Comparison

`Hybrid` shows consistent, incremental gains on each of the six models evaluated as the extraction call (Figure 3). The average gain from the provisional answer to the final answer was +0.07, with `Luna` showing the largest gain of +0.10. `Purist` F1 closely tracks general medical capability as measured by Artificial Analysis’ Healthcare and Medical Index. The models cluster into three groups: strong (`Gemini` 0.86, `Grok` 0.85), medium (`DeepSeek` 0.82, `Luna` 0.79) and weak (`Qwen` 0.76, `Gemma` 0.72), which largely reflect model size and cost.

Table VIII presents a wider view of performance: how often did the models provide exact evidence and adhere to the exact schema requirements. The four hosted models quoted an exact substring of the letter as evidence for the selected event in 99% or more of letters; the two local models paraphrased more often (`Qwen` 97%, `Gemma` 93%) and needed more non-semantic format repair. An exact span is not proof of a correct answer, but it is proof of a verifiable answer. Overall, the models demonstrated they are highly capable of generating rich schemas and following complex task instructions consistently and reliably.

Table IV also lists the two earlier papers that used the same `pragmatic` metric on different data. `Holgate et al.` [8] ran `Llama 2` on local hospital servers with eight Nvidia V100 GPUs and reached a `pragmatic` F1 of 0.73 without fine-tuning. In `Hybrid`, `Qwen` reached a `pragmatic` F1 of 0.80 running locally on a laptop with a single GPU. This shows that the design can execute on local hardware under the same synthetic task conditions.

E. Model Configuration

Raising the reasoning effort level did not improve performance. `Gemini`’s `purist`-F1 score was 0.86 at low and 0.84 at high; a paired difference of +0.02, 95% CI -0.01 to 0.04; $p = 0.31$. Extra reasoning adds cost and latency without improving

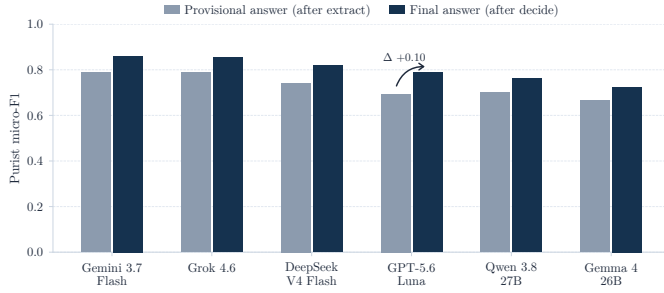


Fig. 3: Hybrid Purist F1 for six models: the LLM’s provisional answer after extract and the final answer after rule-based decide. Every model gains at decide; Luna gains most (0.69 to 0.79).

TABLE VIII: Six-model comparison: performance, exact evidence and schema adherence

Model	Purist	Prag.	Exact ev.	Repairs	Unparsed
Gemini 3.7 Flash	0.86	0.88	99.8%	0	1
Grok 4.6	0.85	0.89	100%	0	0
DeepSeek V4 Flash	0.82	0.85	99.6%	2	0
GPT-5.6 Luna	0.79	0.82	98.9%	0	1
Qwen 3.8 27B (local)	0.76	0.81	96.7%	3	2
Gemma 4 26B (local)	0.72	0.77	92.9%	15	8

Exact ev.: the quoted span for the selected event is an exact substring of the letter. Repairs: letters needing JSON dialect repair. Unparsed: letters with no valid record after repair, scored as incorrect.

extraction. The temperature setting also had no significant effect: 0.86 at temperature 0 versus 0.84 at temperature 1 (paired difference +0.02, 95% CI -0.01 to 0.04; $p = 0.20$). The task is tightly defined by an explicit schema, which limits the expected variability by design and partly explains why neither setting mattered.

F. Error Analysis

The worst performing category was unknown with an F1 score of 0.74. 69% of the errors (37/54) are incorrect answers of unknown. The full error distribution is shown in the confusion matrix (Figure 4).

The three most common errors are infrequent $\rightarrow$ unknown (16), frequent $\rightarrow$ unknown (13), and seizure free $\rightarrow$ unknown (8). The errors reflect a cautious approach when the evidence is ambiguous, for example the model recognised the phrase ‘these episodes crop up most shifts’ as the primary seizure frequency description but refused to make an assumption about how many shifts the person works. The gold-standard label was multiple per week but the model chose unknown. This cautious assessment is justifiable, and because the exact evidence is preserved, downstream users can see the proposed answer and draw their own conclusion from the quoted span.



Fig. 4: Hybrid confusion matrix on the pragmatic categories (test set). Most errors are incorrect predictions of unknown.

TABLE IX: Experimental environment

Component	Setting
Local models	Qwen 3.8 27B and Gemma 4 26B served by Ollama 0.32.15 on a Dell XPS 16 9640 (Windows 11, Intel Core Ultra 9 185H, 32 GB RAM, NVIDIA RTX 4070 Laptop GPU, 8 GB VRAM)
Hosted models	Gemini 3.7 Flash, Grok 4.6, GPT-5.6 Luna and DeepSeek V4 Flash, called from a Mac mini M1 (16 GB, macOS 26.5.2) directly (OpenAI, DeepSeek) or via OpenRouter and AI Gateway
Model settings	Temperature 0 (Luna accepts only 1); thinking or effort level low.
Software	Python 3.14.4, DSPy 3.2.1, Pydantic 2.13.4, LiteLLM 1.87.1

G. Experimental Environment

Table IX summarises the hardware, software and model settings used. DSPy co-ordinated the model calls and required structured JSON output, Pydantic validated each record against the schema, and LiteLLM delivered API calls to Ollama and the model routers. Full specifications, API settings and replay artefacts are in the supporting materials.

V. DISCUSSION

This paper shows that, on the Gan synthetic benchmark, one prompted LLM call and a decision policy extract an epilepsy patient’s current seizure frequency pattern at least as accurately as the fine-tuned LLM previously reported on a different held-out sample of the same corpus [3]. A single prompt designed to extract every candidate with exact evidence, followed by rules or a second call that decide from that evidence, matched that benchmark. The same design also executed on a laptop-class GPU with open-weight models, at a lower but still competitive score. Later researchers or healthcare organisations can evaluate it on their own data and hardware; it is not ready for local deployment.

The method is particularly effective at identifying the categories that mark the greatest disease burden. F1 scores of 0.92 for daily and 0.91 for more than weekly are around 0.2 higher than the category scores previously reported [3], although on a different sample. Structured seizure frequency of this kind is intended for later research uses: cohort identification for trials, retrospective and longitudinal analyses of treatment response, and outcome variables for predictive models [13], which together can widen the scope of real-world evidence [17]. Those are intended uses, not capabilities demonstrated here; each requires validation on real records first.

Past work has typically tasked the LLM with producing a single answer in a fully normalised form [3]. That discards the semantic depth of the narrative, which clinicians value because it conveys reasoning, uncertainty and vague impressions, and those nuanced observations are more accurate and reliable [18]. Current LLMs can populate a richer record that preserves them: every candidate with its quoted span, the policy applied, and the label it produced. The ablations show that this record is also what the accuracy depends on. A more explainable system is often a more useful system in a clinical setting [14], and here it is not a less accurate one.

There are several limitations. Synthetic data was used for all evaluations, and while past research has shown models fine-tuned on this dataset generalise effectively to real patient data [3], the methods tested here remain untested on real letters; no claim is made about clinical validity, workflow fit or privacy compliance. The held-out samples differ. Past research has shown that rules tuned on one corpus can generalise poorly to others, so the gains from rules may not transfer to other datasets [6]. The stage scores are cumulative stops rather than isolated stage metrics. Evidence quality was measured only as exact-substring adherence, which does not show that a span semantically supports the label; the gold standard’s single reference quote is a loose comparator for that purpose, since it may be paraphrased or abbreviated. A development-set study of semantic evidence sufficiency, adjudicated from the extracted spans and the policy alone, is specified as the next step before any further held-out evaluation. Finally, the method was applied to one narrow task. The same method has been configured for a multi-fact epilepsy inventory task [4], to be reported separately, and future work should evaluate whether the design generalises to other clinical IE tasks and whether a more general framework, in which an LLM chooses how to represent facts and which policy to apply, adds value as reported in oncology [19] and cardiology [20].

VI. AI ETHICAL CONSIDERATIONS

This evaluation used only the synthetic clinic letters released by Gan *et al.* [3]. Those letters are modelled on real UK epilepsy letters but do not contain identifiable patient records. Hospitals prefer processing patient data on their own servers as it is highly sensitive [5]. Two of the six models were run entirely on local hardware to test whether the design can execute in such a setting on synthetic data, not as a privacy-compliance assessment.

LLMs can invent evidence, so careful consideration was given to the prompt instructions and schema design. The model is required to quote exact evidence spans from the letter, and every later change to the answer is made by a named rule or a second call applied to that quoted evidence. A reviewer can therefore trace any final label back to the text that supports it and to the policy that decided it [14]. This transparency matters more than headline accuracy in a clinical setting, where an unexplained answer cannot be safely acted on.

The letters contain no demographic attributes, so unequal performance across patient groups cannot be tested, and the synthetic corpus may not reflect the language of every clinic. Bias can enter through the models, the prompts and the decision rules; for example, the cautious default to `unknown` (Section IV-F) would systematically under-count patients whose seizures are described vaguely. Rules make the decision policy inspectable and changeable by local clinicians, but they do not remove the need for local clinical review and fairness testing on real records before any operational use.

VII. CONCLUSION

Seizure frequency is a core outcome measure in epilepsy, recorded almost exclusively in narrative clinic letters. This paper proposes a pipeline in which one LLM call extracts every candidate fact with its quoted evidence in a canonical form, then a decision applies a fixed policy to those candidates.

Both Hybrid (Purist F1: 0.86) and LLM-only (0.85) exceeded the benchmark previously reported for a fine-tuned LLM (0.81) on a different held-out sample of the same synthetic corpus. The provisional answer reaches 0.79; applying the decision policy to the evidence it found adds a further +0.06 to +0.07, whether that policy is applied by rules or by a second LLM call. The result depends on the extraction prompt’s examples, closed label forms and evidence obligation, and it held across six models, including two run on a single laptop GPU.

Automatically extracting seizure frequency patterns from EHRs could make this outcome measure available for large-scale clinical research: cohort identification, retrospective studies and predictive modelling. Future research should evaluate it on real patient data and on a broader set of clinical information extraction tasks before any of those uses.

REFERENCES

- [1] S. Wigglesworth, A. Neligan, J. M. Dickson, A. Pullen, E. Yelland, T. Anjuman, and M. Reuber, “The incidence and prevalence of epilepsy in the United Kingdom 2013–2018: A retrospective cohort study of UK primary care data,” *Seizure*, vol. 105, pp. 37–42, 2023.
- [2] R. S. Fisher *et al.*, “ILAE official report: A practical clinical definition of epilepsy,” *Epilepsia*, vol. 55, no. 4, pp. 475–482, 2014.
- [3] Y. Gan *et al.*, “Reproducible synthetic clinical letters for seizure frequency information extraction,” arXiv:2603.11407, 2026.
- [4] B. Fonferko-Shadrach *et al.*, “Annotation of epilepsy clinic letters for natural language processing,” *J. Biomed. Semantics*, vol. 15, no. 17, 2024.
- [5] B. Holgate, J. Davies, S. Fang, J. S. Winston, J. T. Teo, and M. P. Richardson, “Fine-tuning LLMs to extract epilepsy seizure frequency data from health records,” in *Proc. 24th Workshop on Biomedical Language Processing (BioNLP)*, 2025, pp. 44–55.

- [6] B. M. Decker *et al.*, “Development of a natural language processing algorithm to extract seizure types and frequencies from the electronic health record,” *Seizure*, vol. 101, pp. 48–51, 2022.
- [7] K. Xie *et al.*, “Extracting seizure frequency from epilepsy clinic notes: a machine reading approach to natural language processing,” *J. Amer. Med. Informat. Assoc.*, vol. 29, no. 5, pp. 873–881, 2022.
- [8] B. Holgate, S. Fang, A. Shek, M. McWilliam, P. Viana, J. S. Winston, J. T. Teo, and M. P. Richardson, “Extracting epilepsy patient data with Llama 2,” in *Proc. 23rd Workshop on Biomedical Natural Language Processing (BioNLP)*, 2024, pp. 526–535.
- [9] K. Xie, B. Litt, D. Roth, and C. A. Ellis, “Quantifying clinical outcome measures in patients with epilepsy using the electronic health record,” in *Proc. BioNLP Workshop*, 2022, pp. 369–375.
- [10] M. Fernandes, A. Cardall, L. M. V. R. Moura, C. McGraw, S. F. Zafar, and M. B. Westover, “Extracting seizure control metrics from clinic notes of patients with epilepsy: A natural language processing approach,” *Epilepsy Res.*, vol. 207, 2024, Art. no. 107451.
- [11] T. Brown *et al.*, “Language models are few-shot learners,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1877–1901.
- [12] M. Agrawal, S. Hegselmann, H. Lang, Y. Kim, and D. Sontag, “Large language models are few-shot clinical information extractors,” in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2022, pp. 1998–2022.
- [13] L. Cui, S. S. Sahoo, S. D. Lhatoo, G. Garg, P. Rai, A. Bozorgi, and G.-Q. Zhang, “Complex epilepsy phenotype extraction from narrative clinical discharge summaries,” *J. Biomed. Informat.*, vol. 51, pp. 272–279, 2014.
- [14] S. Fu *et al.*, “Clinical concept extraction: A methodology review,” *J. Biomed. Informat.*, vol. 109, 2020, Art. no. 103526.
- [15] Y. Wang *et al.*, “Clinical information extraction applications: A literature review,” *J. Biomed. Informat.*, vol. 77, pp. 34–49, 2018.
- [16] L. Chiticariu, Y. Li, and F. Reiss, “Rule-based information extraction is dead! Long live rule-based information extraction systems!,” in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2013, pp. 827–832.
- [17] R. E. Sherman *et al.*, “Real-world evidence—what is it and what can it tell us?” *N. Engl. J. Med.*, vol. 375, no. 23, pp. 2293–2297, 2016.
- [18] S. T. Rosenbloom, J. C. Denny, H. Xu, N. Lorenzi, W. W. Stead, and K. B. Johnson, “Data from clinical notes: A perspective on the tension between structure and flexible documentation,” *J. Amer. Med. Informat. Assoc.*, vol. 18, no. 2, pp. 181–186, 2011.
- [19] Z. Yang *et al.*, “CLINES: Clinical LLM-based information extraction and structuring agent,” medRxiv:10.64898/2025.12.01.25341355, 2025.
- [20] J. Wang *et al.*, “Scaling clinician-grade feature generation from clinical notes with multi-agent language models,” arXiv:2508.01956, 2025.
- [21] D. Xu, M. Gopale, J. Zhang, K. Brown, E. Begoli, and S. Bethard, “Unified Medical Language System resources improve sieve-based generation and Bidirectional Encoder Representations from Transformers (BERT)-based ranking for concept normalization,” *J. Amer. Med. Informat. Assoc.*, vol. 27, no. 10, pp. 1510–1519, 2020.